Data Journalism

Session 6: Mapping and geodata

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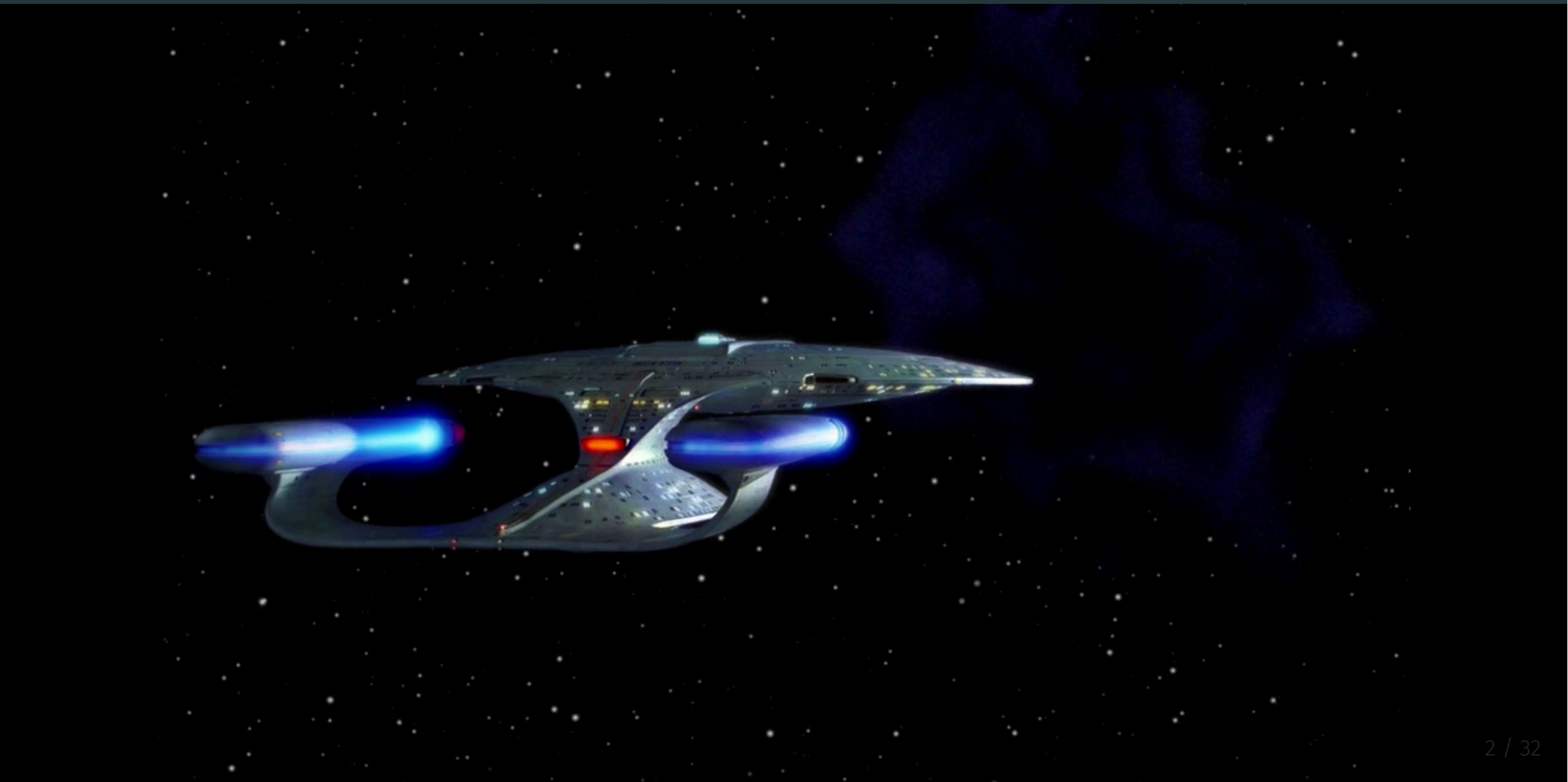
Hertie School | GRAD-E1493

Space. The final frontier...



Today

1. Why space matters for journalism
2. Maps: dos and don'ts
3. Beyond maps

Why space matters for journalism

Why space matters generally

Everything is related to everything else,
but near things are more related than distant things.

Tobler, Waldo. (1970). A Computer Movie Simulating Urban Growth in the Detroit Region. *Economic Geography* 46:234-240. https://doi.org/10.2307/143141

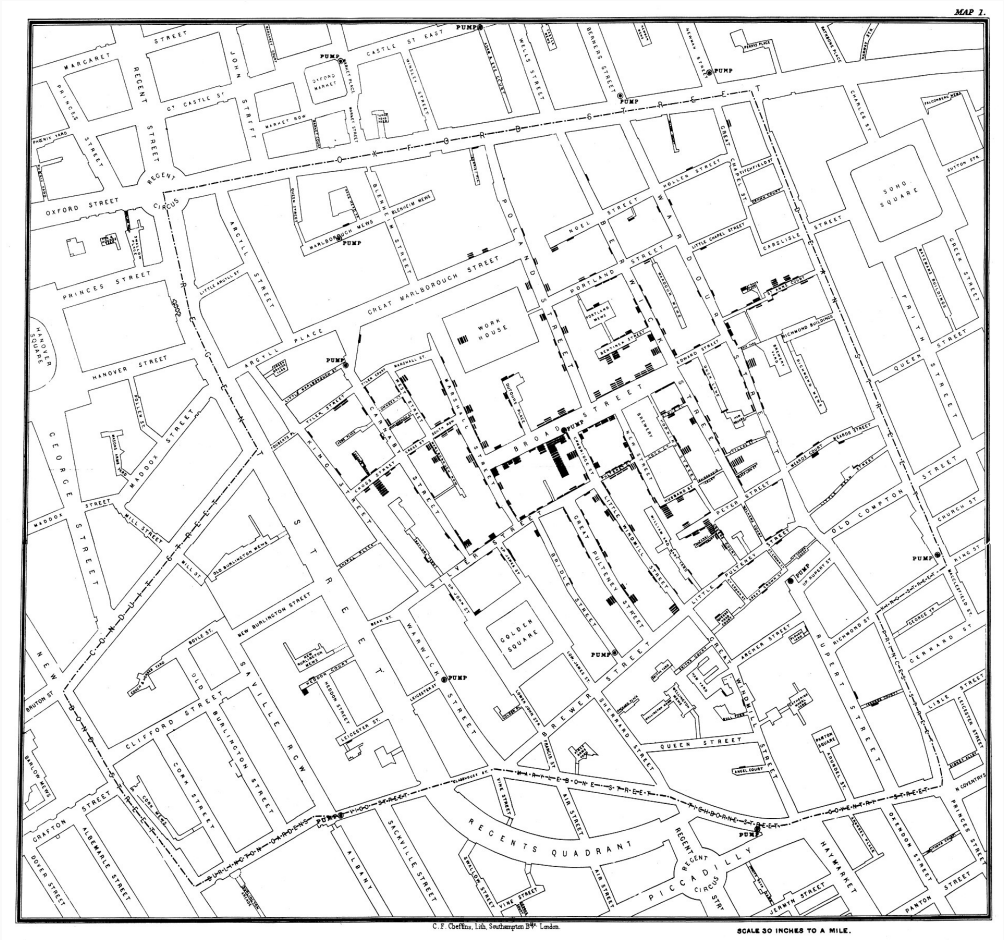
Why space matters for journalism

The "where" of a story is central to journalism.

Gomes, A., Brito, E., Morais, L. A., & Ferreira, N. (2025). How Do Data Journalists Design Maps to Tell Stories? *IEEE Transactions on Visualization and Computer Graphics*.

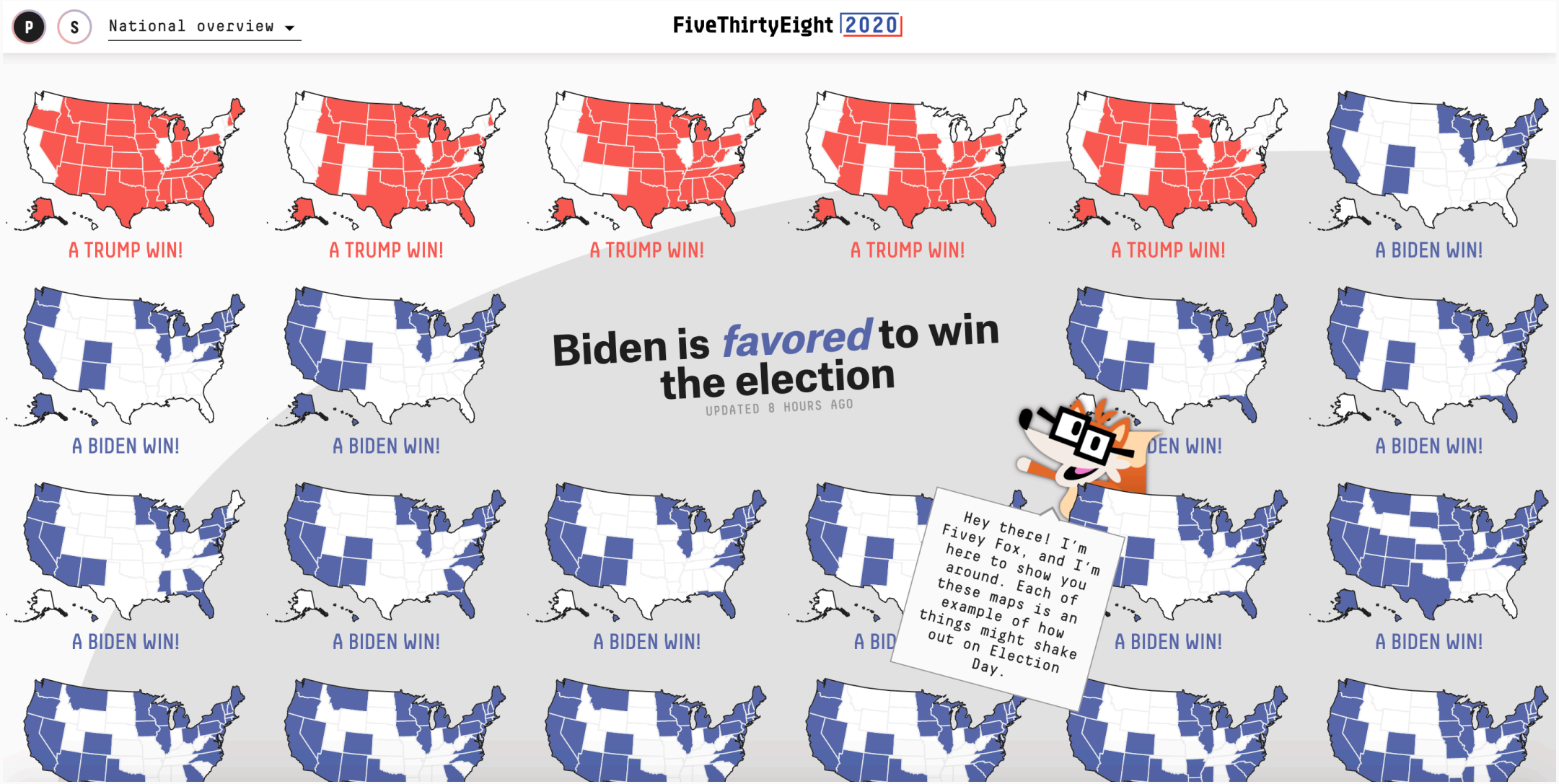
N. Usher. News cartography and epistemic authority in the era of big data: Journalists as map-makers, map-users, and map-subjects. *New Media & Society*, 22(2):247–263, 2020.

1854 Broad Street cholera outbreak



- One of the most famous data viusalizations of all times: John Snow's cholera case map.
- The Broad Street cholera outbreak in in Soho, London in 1854 was studied by physician John Snow to study its causes (rival hypotheses: germ-contaminated water vs. airborne transmission).
- The germ theory was not established at this point but the map helped highlight how cases clustered around a contaminated pump (which was by far not the only source of contaminated water though).
- Fun fact: this is what doing good data viz gives you:

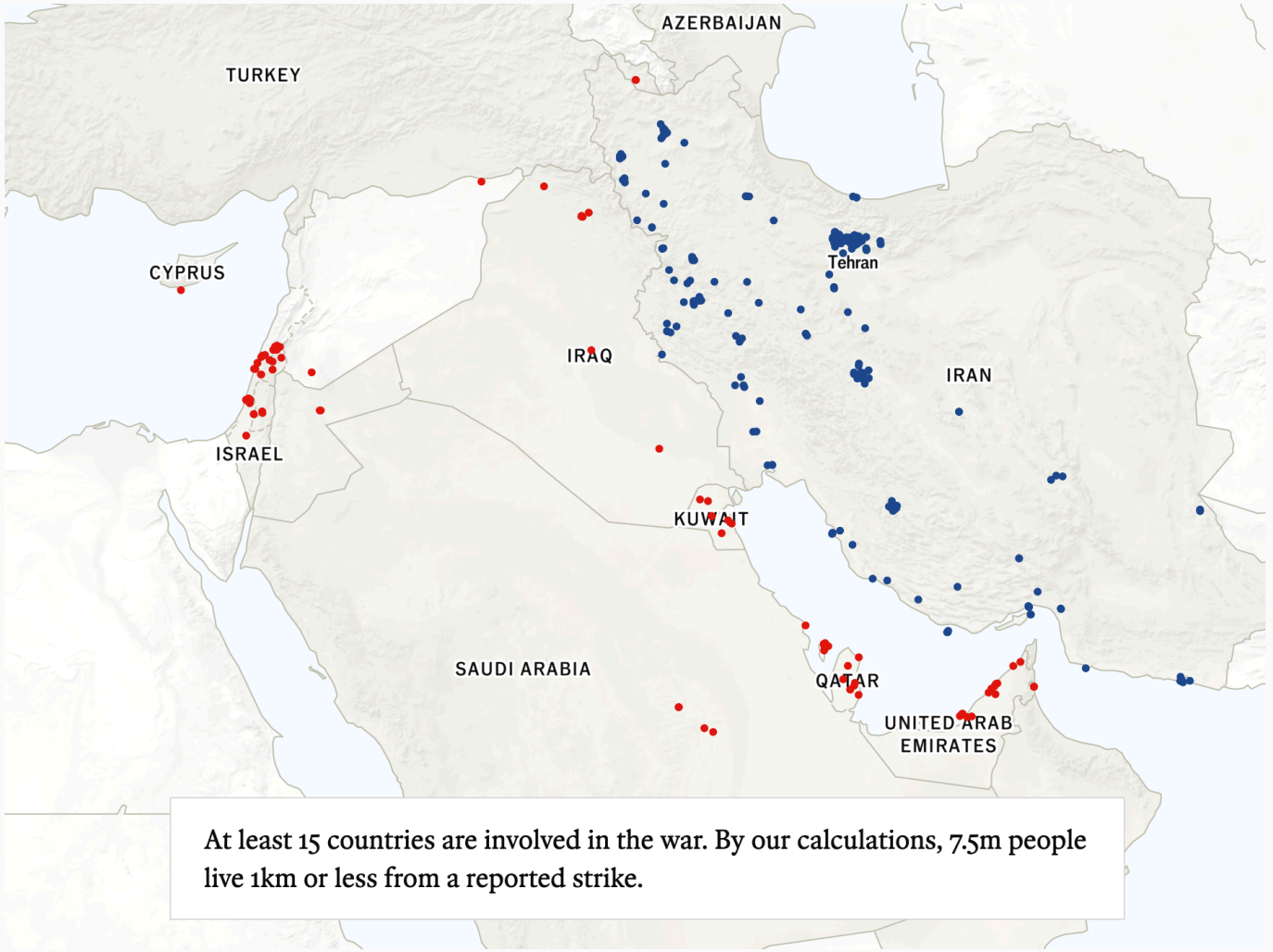
Mapping political outcomes



Mapping political outcomes

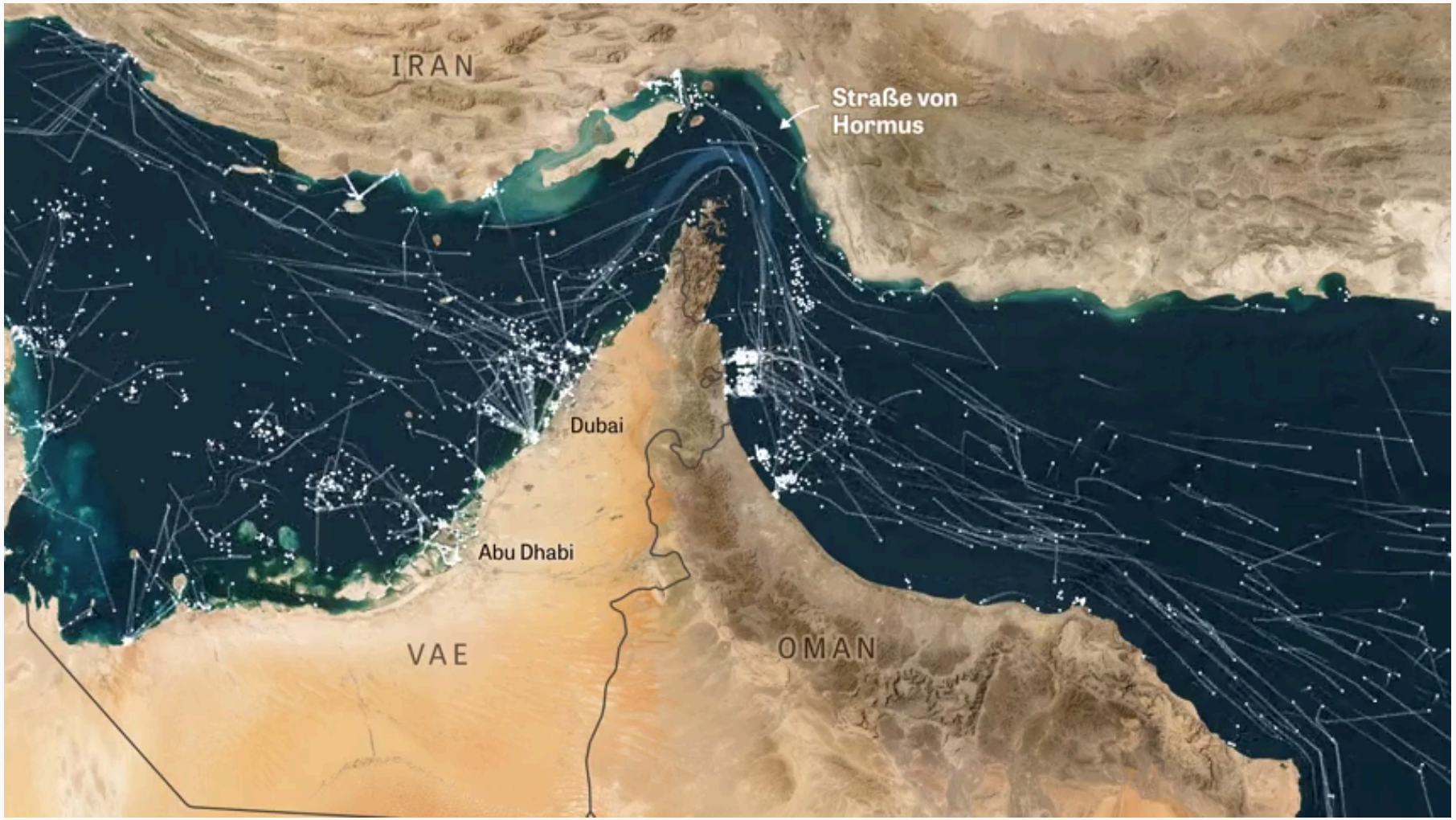
Source ZEIT Online

Unterstanding international affairs



Source The Economist

Unterstanding international affairs



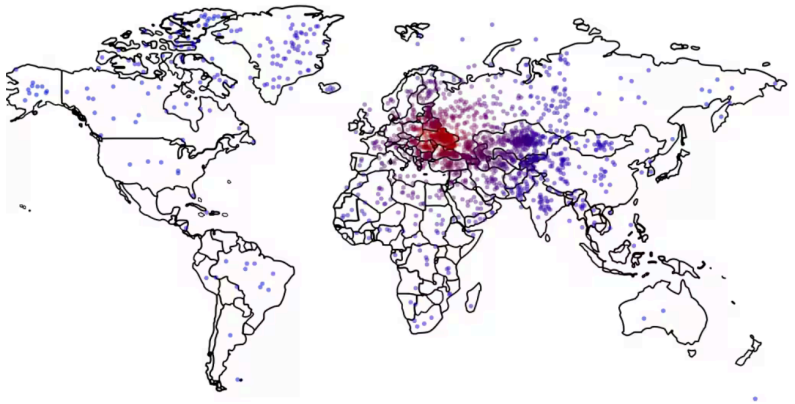
`Source` Die Zeit

Educating geographically illiterate audiences

Monkey Cage

The less Americans know about Ukraine's location, the more they want U.S. to intervene

April 7, 2014 More than **11 years ago**



Where's Ukraine? Each dot depicts the location where a U.S. survey respondent situated Ukraine; the dots are colored based on how far removed they are from the actual country, with the most accurate responses in red and the least accurate ones in blue. (Data: Survey Sampling International; Figure: Thomas Zeitzoff/The Monkey Cage)

By Kyle Dropp, Joshua D. Kertzer and Thomas Zeitzoff

`Source` Washington Post/Monkey Cage; see also Good Authority.

The design space of journalistic maps

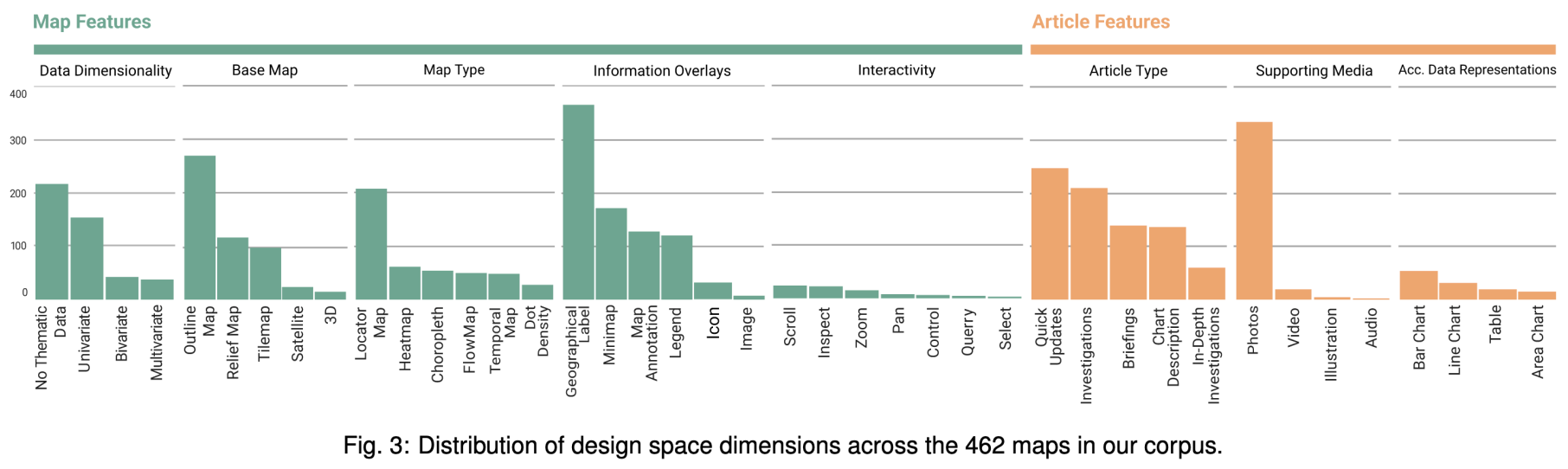
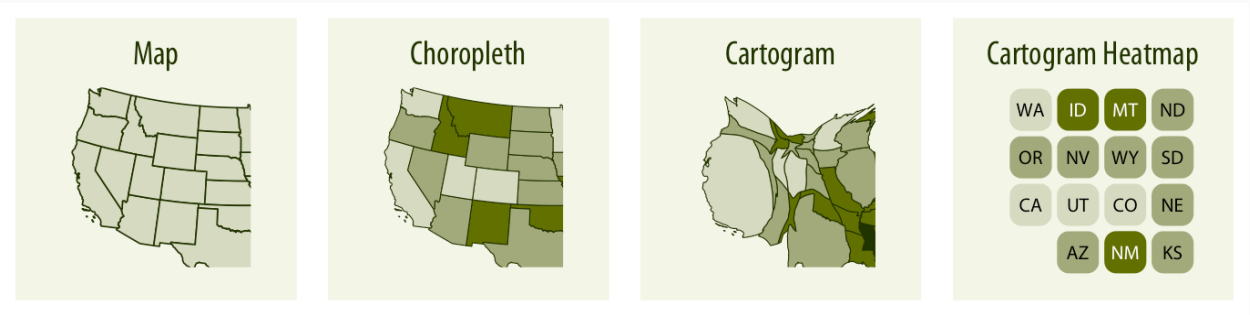


Fig. 3: Distribution of design space dimensions across the 462 maps in our corpus.

`Source` Gomes et al. (2025) "How do Data Journalists Design Maps to Tell Stories?"



`Source` Wilke, Fundamentals of Data Visualization

Maps: Dos and don'ts

Fox News 2009

Source

WSJ 2012

OPINION

Obama's Divided Nation

ama presides over
ica more divided
any time in 50
ry that was riven
g racial lines gath-
2008 to elect its
ident. That presi-
four years dividing
the basis of eco-
The campaign re-
vealed no evi-
dence that Mr.
Obama will close
the chasm he has
created between
his voters and
those he attacked
and vilified.

It may be true
that Mitt Romney
failed to respond

drawn attention to what hap-

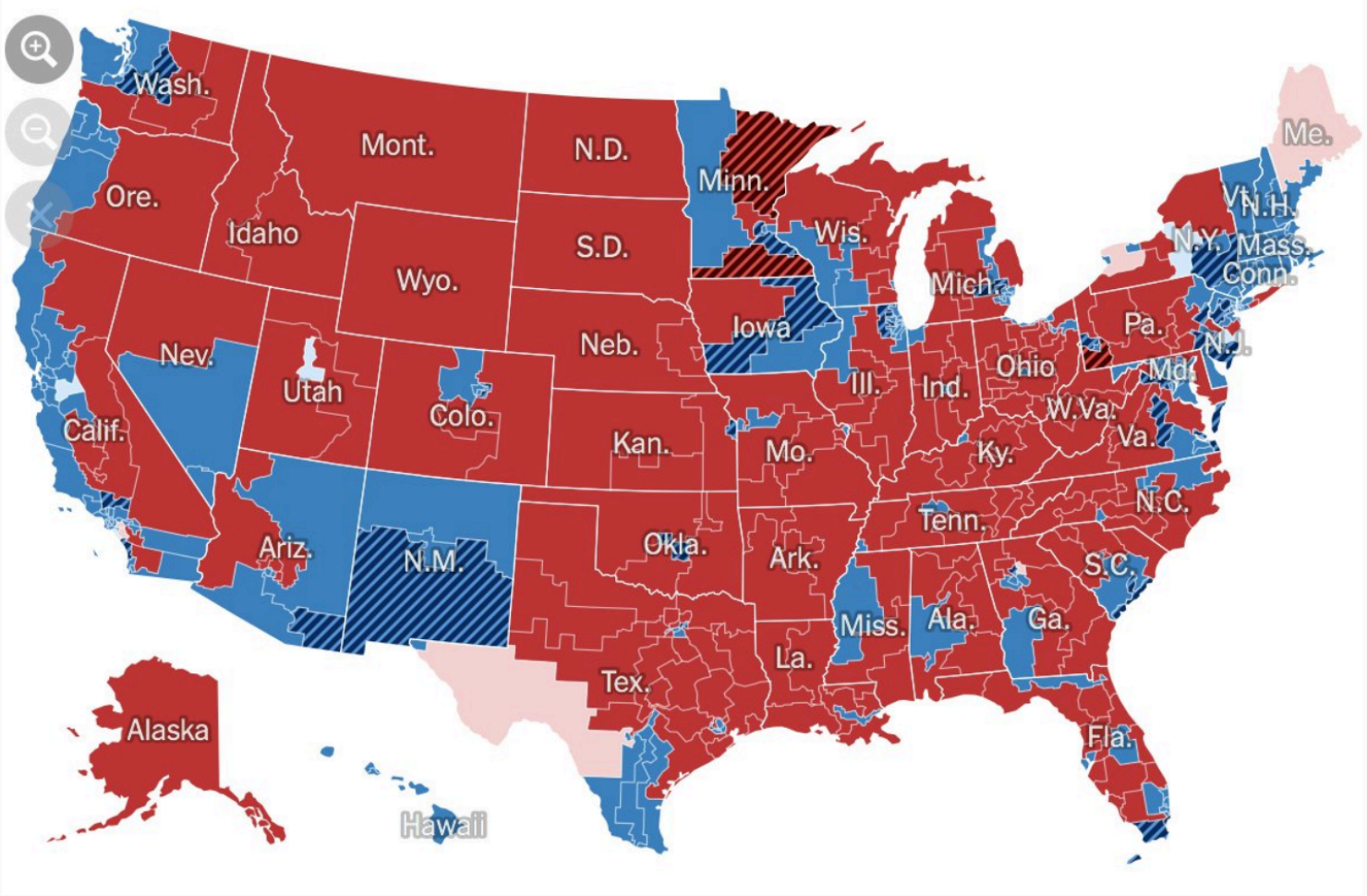
Obama spokestotum replied:

problem with pols, e
verbally facile as Mr.
that in crunch time,
reverts to No. 1. Exi
that 9% of the electo
who to vote for jus
Tuesday; and amo
42% said Mr. Oban
Sandy response—t
tie photo-op—wa
factor. Of those,
voted for Mr. Ob
Mr. Christie is o
politico who is

Yes, Republic
across two pres
that there are
how crudely
issue like ill
Blowing up th
if you thoug
day's results

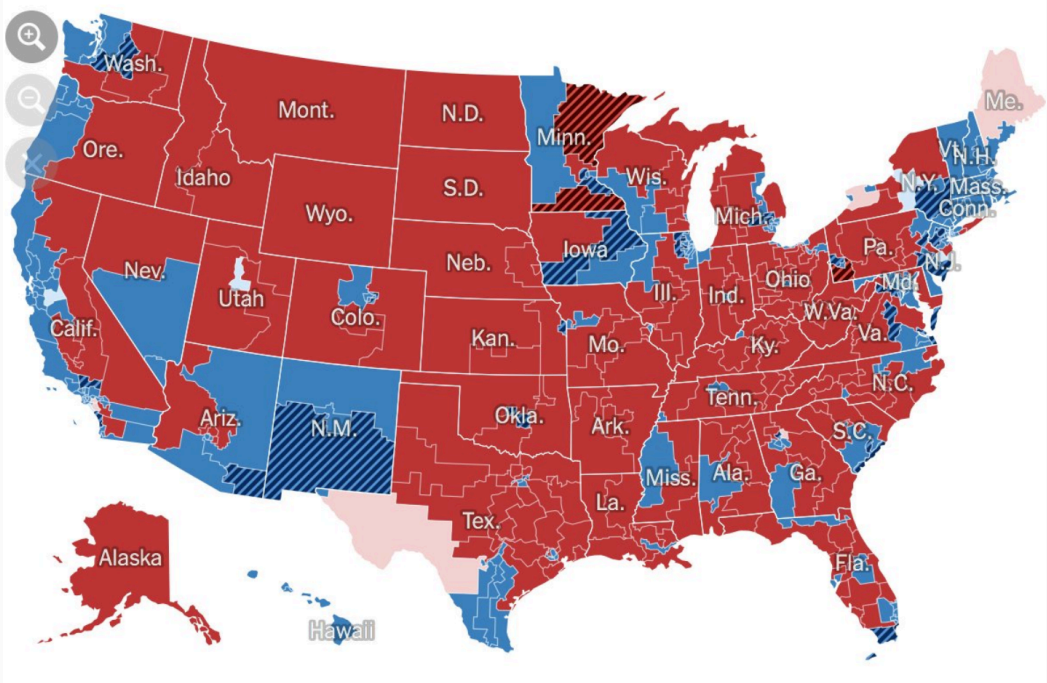
Source

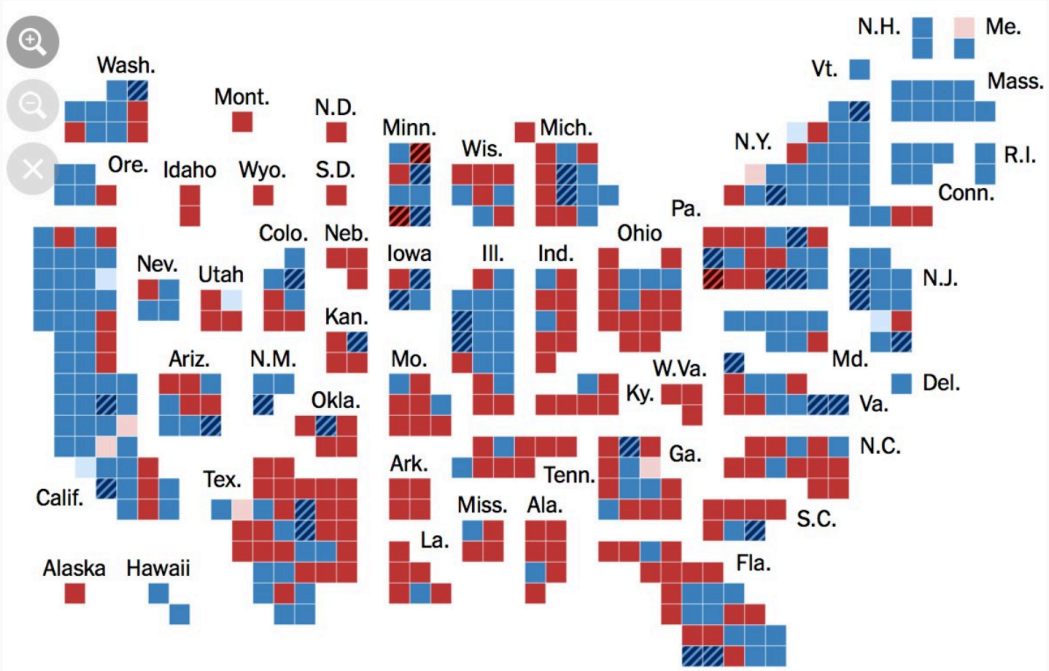
Maps are beautiful ...



2019 US Midterm Election results; source: nytimes.com

… but sometimes misleading

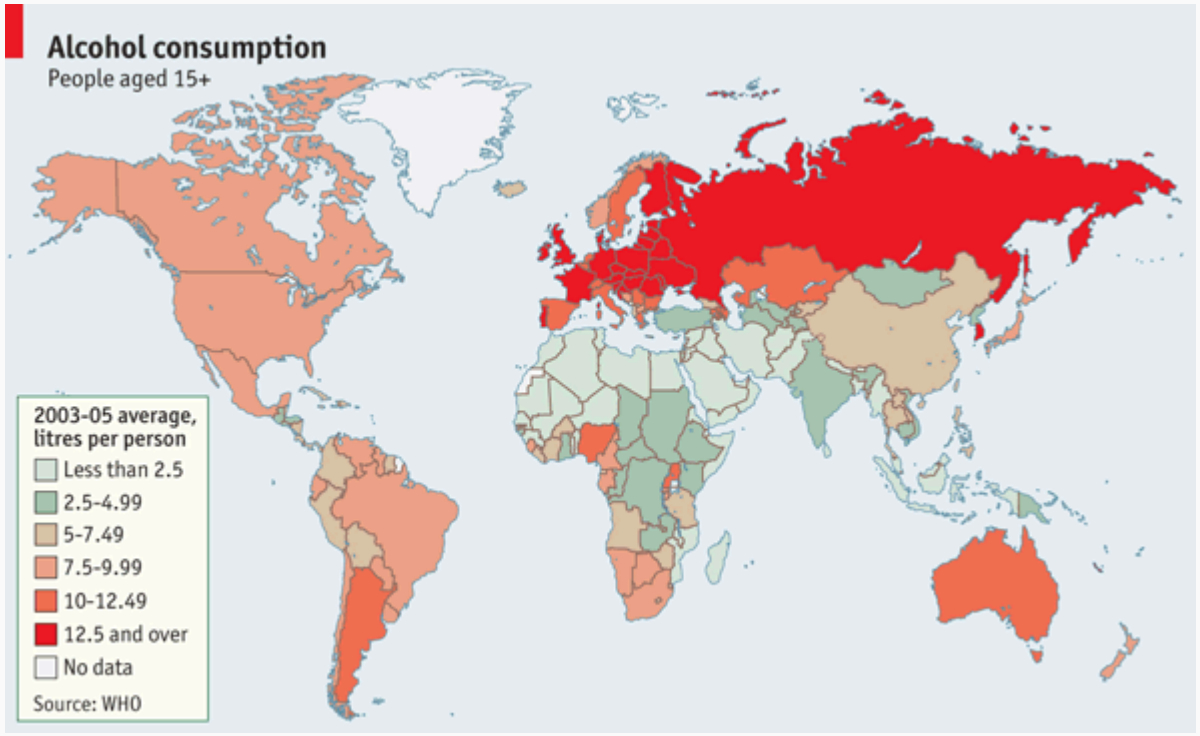


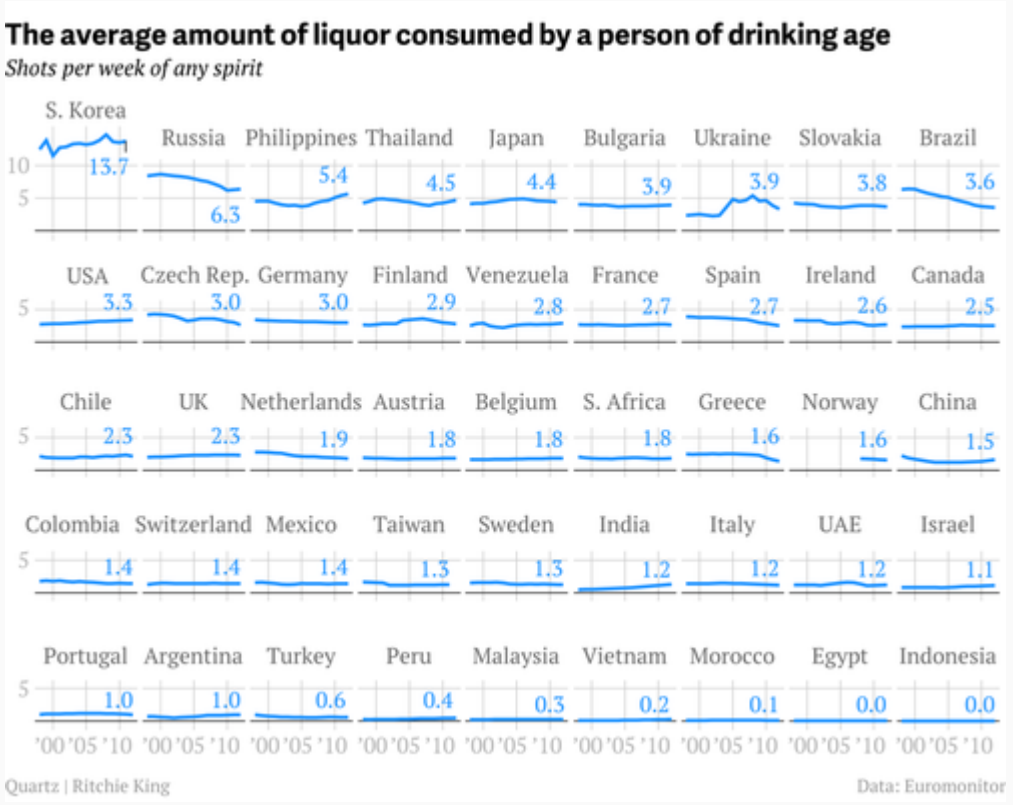


Don't confuse area size with actual data of interest

Large geographic areas attract attention regardless of their data values. Consider cartograms, hex tile maps, or dot density maps to counteract this.

Consider if a map is the right choice





Maps help highlight geographical patterns, but is that your story?

Also discussed here. The small multiple plot on the right does a much better job at showing the trends over time and comparing across countries, while the map on the left is more visually striking but less informative. The map also exaggerates the differences between countries due to the varying sizes of the areas.

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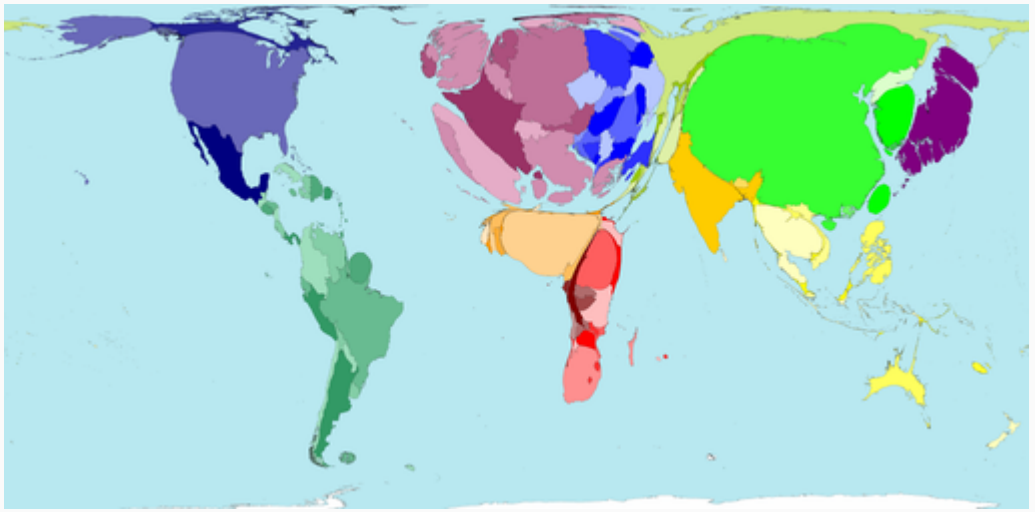

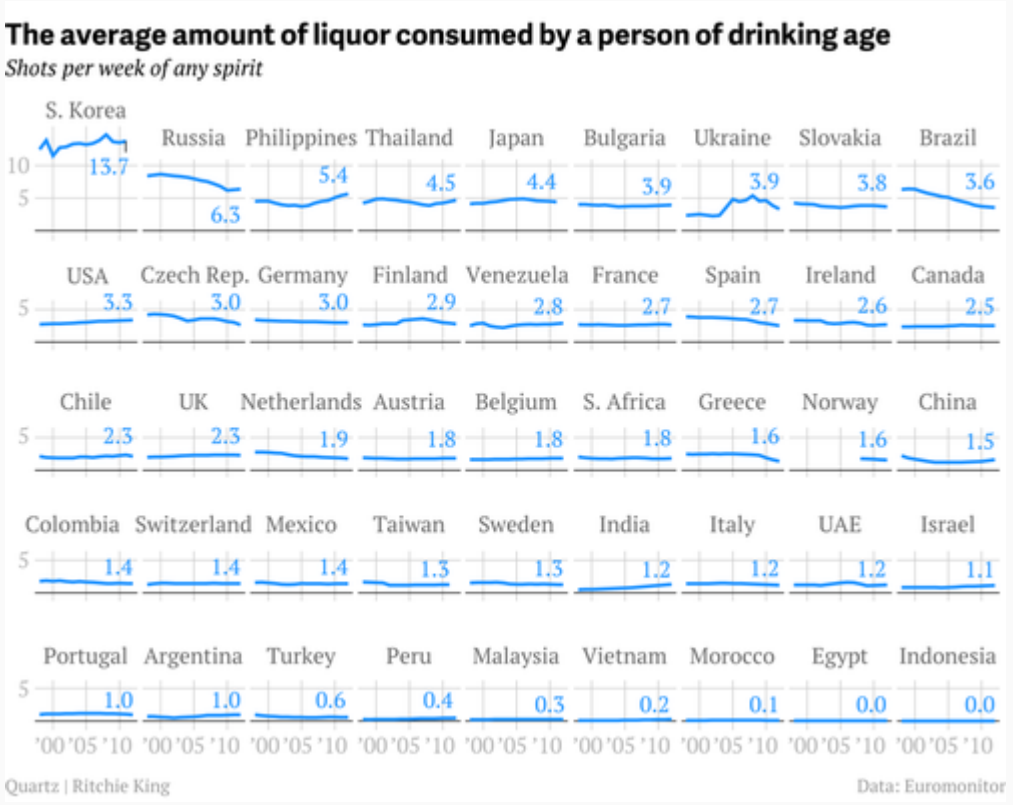

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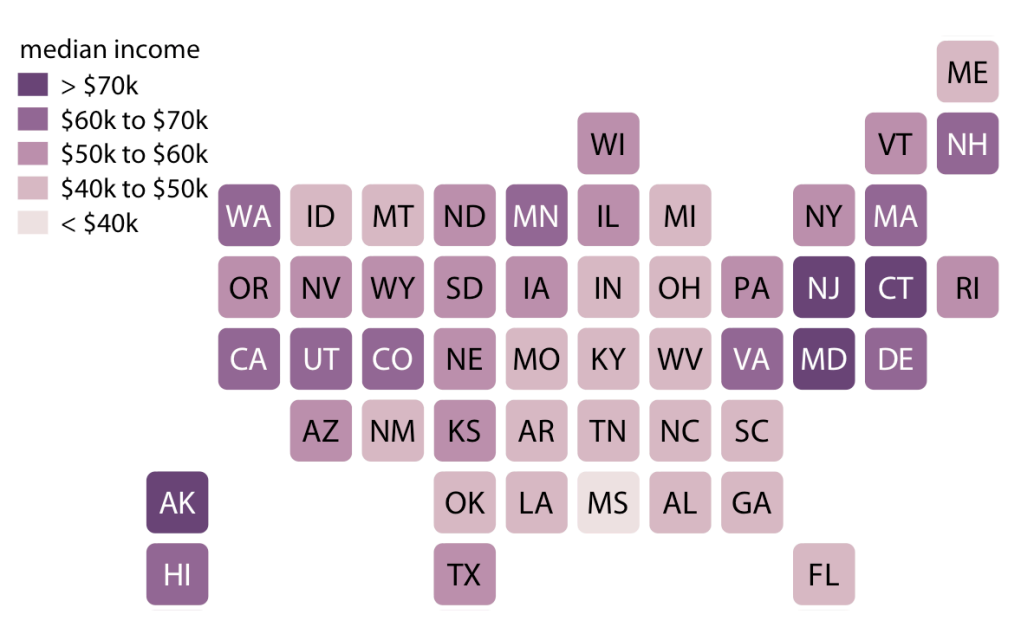
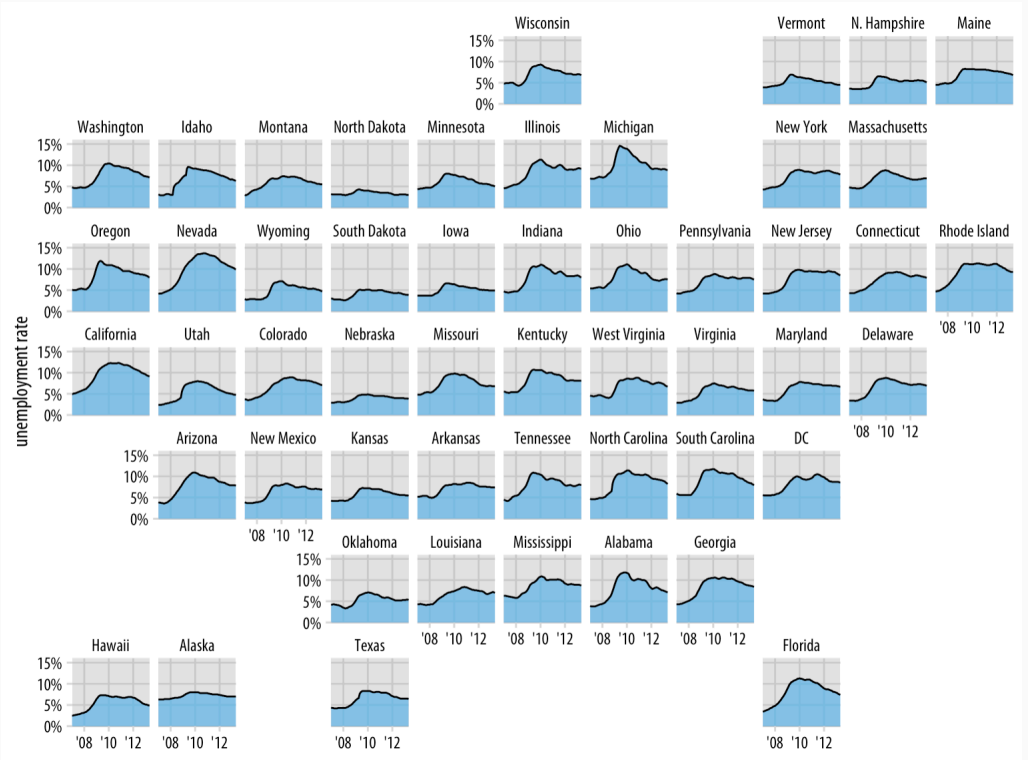
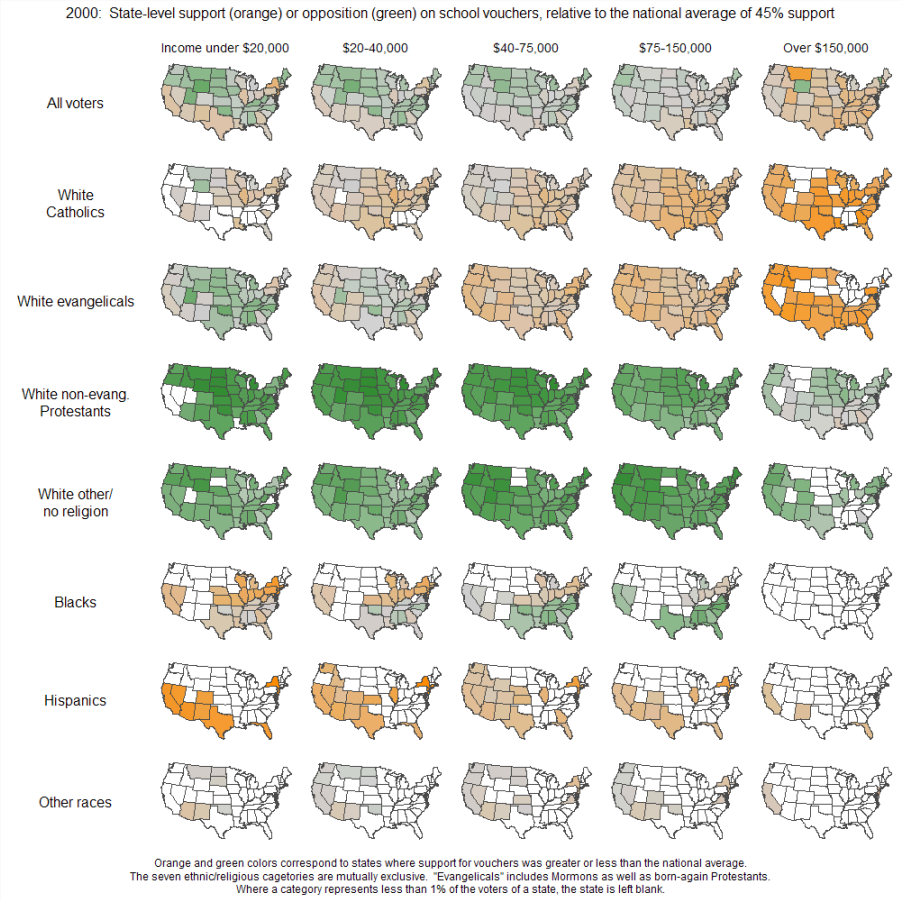
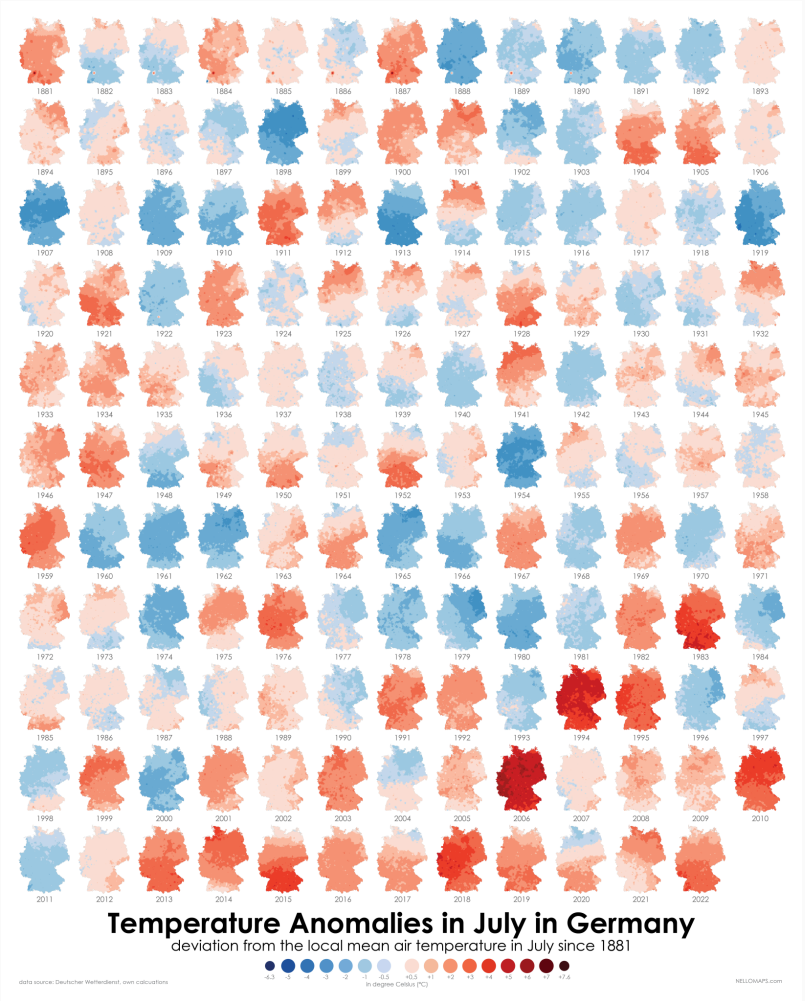
Cartogram heatmaps can help (or confuse?)





Why not try small map multiples?



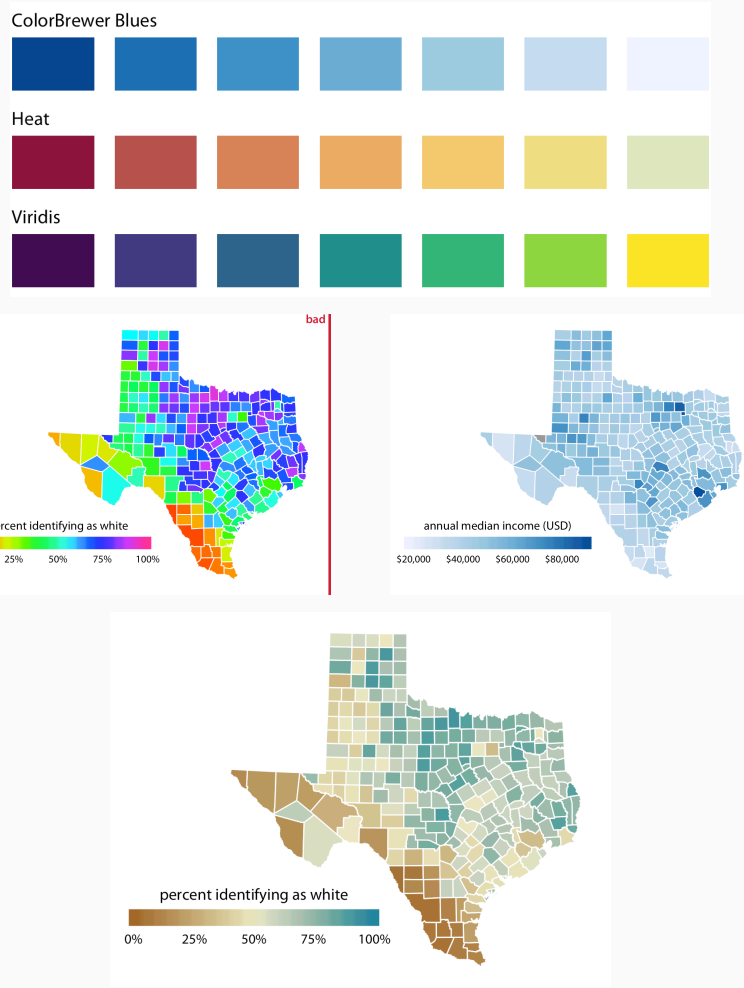


Use color scales wisely

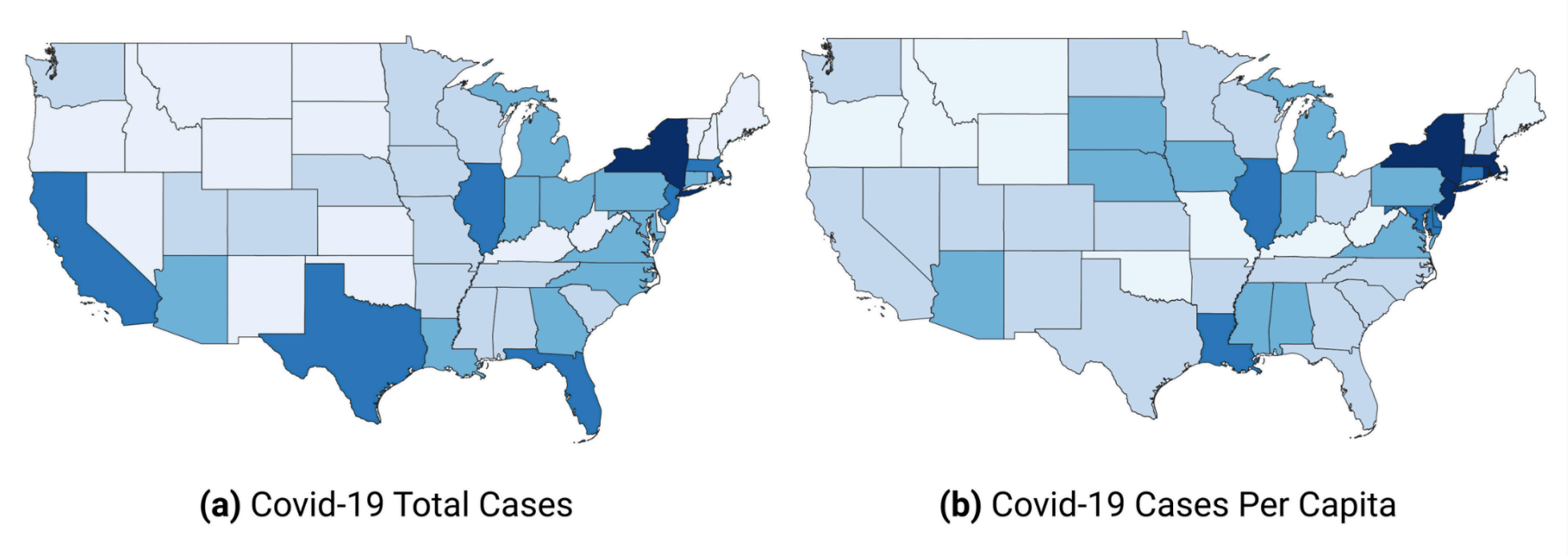
Three fundamental use cases for color in data visualizations:

1. We can use color to **distinguish groups** of data from each other;
2. We can use color to **represent data values**; and
3. We can use color to **highlight**.

Use ColorBrewer to choose palettes.



Normalize your data



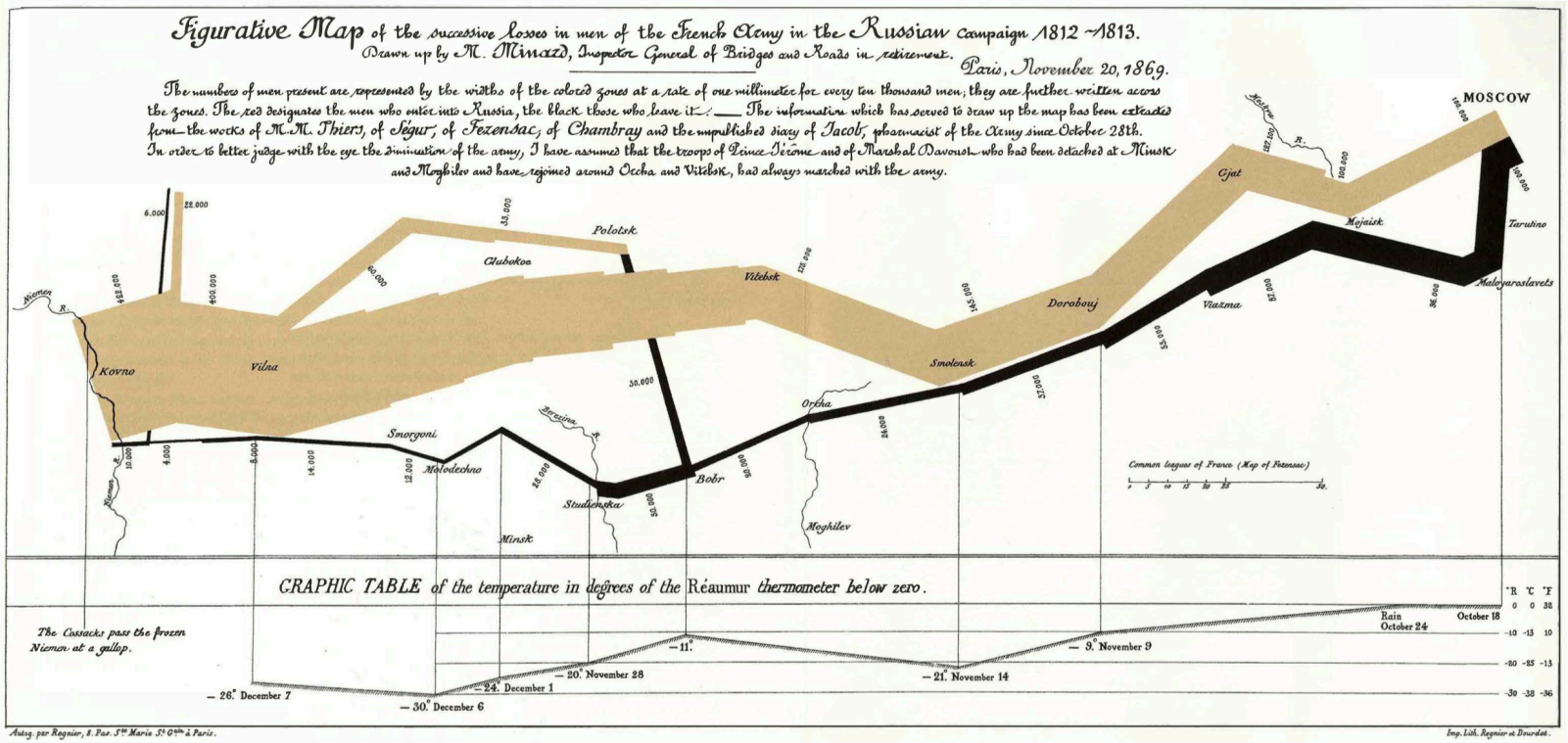
(a) Covid-19 Total Cases

(b) Covid-19 Cases Per Capita

Mapping raw counts in choropleths can be misleading

Absolute counts can be misleading - large units/states dominate (left plot). Instead, normalize using meaningful baselines, such as per capita rates (right plot). If your story is indead about area size, that is probably not necessary. Source.

Keep things simple



Don't overload your map with too many variables

Minard's famous map of Napoleon's march on Russia in 1812 has been regarded a masterpiece of data visualization. It combines multiple variables (geography, time, temperature, and troop size) in a clear and concise way. On the other hand, it's clearly **overloaded**, and one might ask if the same story could not have been told with several plots that are not as busy.

Don't forget about your projection

The issue

The Earth is a sphere, but maps are flat. The choice of projection can affect how readers perceive the data and the story you want to tell.

On a Mercator map, Greenland appears the same size as Africa — but Africa is actually 14× larger. Try thetruesize.com. Use equal-area projections (e.g., Robinson, Equal Earth) for world maps showing data.

`Source` Wilke, Fundamentals of Data Visualization

`Source` Wikipedia

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Source Wilke, Fundamentals of Data Visualization

Source DIE ZEIT

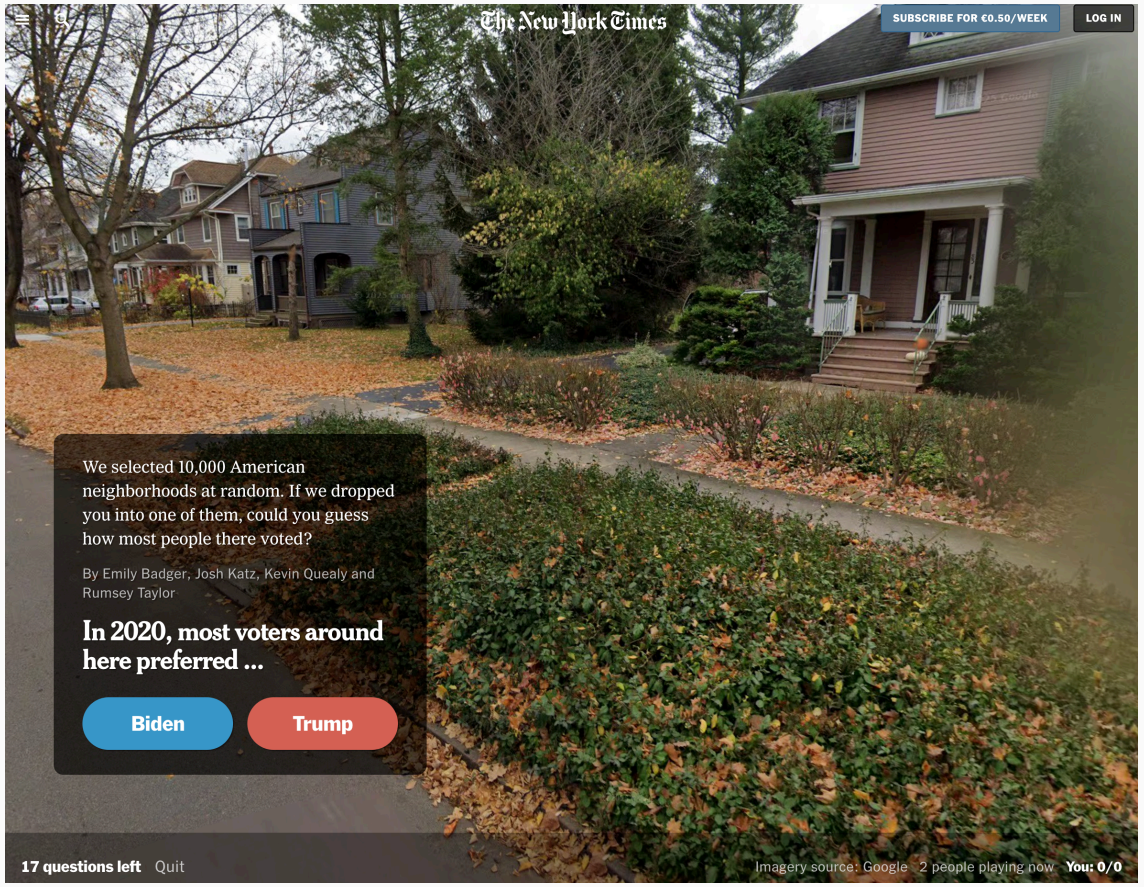
Beyond maps

Discussion

Discuss

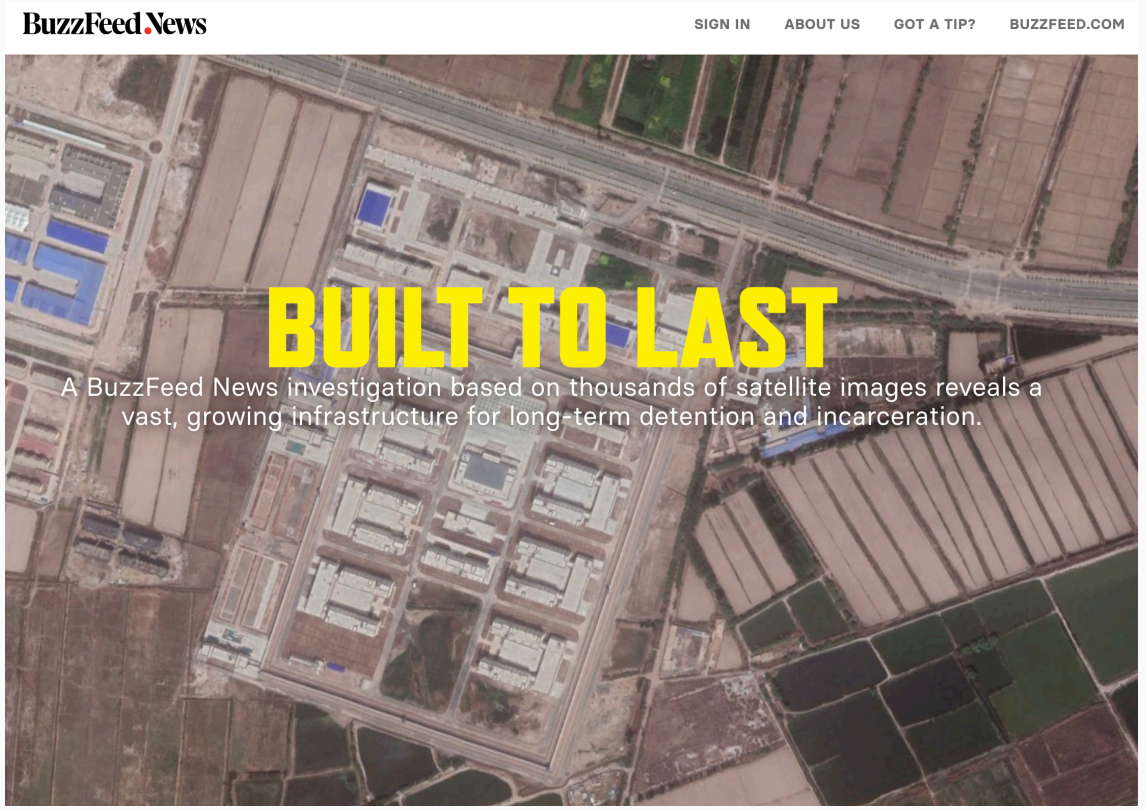
1. **Beyond the map:** Geospatial data is often treated as input for maps — but what other analytical uses does it enable? Think of examples from journalism or your own work.
2. **Spatial thinking as a method:** When does location *explain* something, and when is it just context? How do you know when a spatial analysis adds real insight vs. visual appeal?
3. **Access and limits:** Where does geospatial data come from, and who controls it? What are the ethical and practical constraints journalists face when working with it?

Streetmap data



`Source` New York Times

Satellite data



`Source` Buzzfeed

Geospatial data beyond maps

Location as a join key

Link datasets that share no common ID but share space — e.g., match industrial sites to asthma hospitalization records by proximity.

Distance and proximity as a variable

Not "where is X?" but "how far is X from Y, and does that distance predict something?" — access to schools, pollution exposure, ...

Satellite and remote sensing data as a source

Count ships in a port, measure deforestation, detect construction in sanctioned countries. The imagery is evidence.

Geocoding as a method for record linkage

Convert messy address data into coordinates to deduplicate records or match across datasets that otherwise can't be joined.

> **Coordinates are a universal join key** — they connect datasets that have nothing else in common. That's the analytical superpower, not (just) the map.